

# ATHIRA KRISHNA

Senior ML / AI Engineer

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## PROFESSIONAL SUMMARY

Senior Machine Learning Engineer with 7+ years of experience across Data Science and Data Engineering, delivering scalable AI/ML solutions in healthcare, supply chain, and financial services domains. Expertise in end-to-end machine learning lifecycle including data engineering, feature engineering, model development, deployment, and MLOps. Strong hands-on experience with Python, Machine learning and Deep Learning techniques.

Specialized in Generative AI solutions using Hugging Face Transformers, LangChain, Groq, RAG pipelines, vector databases, prompt engineering, and RAG evaluation frameworks like RAGAS. Skilled in statistical modeling, anomaly detection, time series forecasting, and explainable AI using SHAP. Experienced in building scalable ETL/ELT pipelines with Apache Airflow and implementing CI/CD automation using Jenkins. Strong cloud expertise on Google Cloud technologies including Vertex AI, BigQuery, GCS, and Dataproc. Open to opportunities in Senior Data Science, Machine Learning Engineering, Generative AI, and Data Engineering roles.

## CORE COMPETENCIES

**Generative AI & LLMs:** RAG Pipelines, Transformer Architectures, Prompt Engineering, LangChain, Vector Databases (ChromaDB, FAISS), AI Agents, OpenAI GPT-4, Groq API

**ML & NLP:** Predictive Modeling, Classification, Clustering, Time-Series Forecasting, Deep Learning (CNN, LSTM/RNN), Sentiment Analysis, Dimensionality Reduction (PCA, SVD), SHAP Explainability, Anomaly Detection

**Frameworks & Libraries:** Python, Scikit-Learn, TensorFlow, PyTorch, Pandas, NumPy, PySpark, Hugging Face Transformers, RAGAS

**Data Engineering:** ETL/ELT Pipelines, SQL, Apache Spark (PySpark, Scala), Hadoop, Hive, BigQuery, Feature Engineering, Data Modeling

**Cloud & MLOps:** GCP (Vertex AI, BigQuery, Dataflow, Dataproc, Cloud Composer, GCS), Model Monitoring, CI/CD (Jenkins, GitHub Actions), Apache Airflow, Docker, Kubernetes

**Tools & Visualization:** Streamlit, FastAPI, REST APIs, Qlik Sense, Git, Control-M

## PROFESSIONAL EXPERIENCE

**Senior Data Analyst – Data Science** | Infosys Ltd | Dec 2021 – Oct 2025

Projects: Supply Chain Analytics (OTIF Risk Prediction) | ESG Analytics Platform (Banking) | GenAI R&D

**Project 1: OTIF Delivery Risk Prediction | Supply Chain Analytics** | GCP · Vertex AI · BigQuery · AutoML · Airflow · Python · Scikit-Learn

- Designed and deployed end-to-end ML system to predict OTIF (On-Time In-Full) delivery risk, enabling proactive supply chain delay prevention
- Built binary classification model on large-scale BigQuery data; performed feature engineering and statistical analysis (p-values, hypothesis testing, correlation) to identify key risk drivers — supplier performance, lead time variability, delivery history
- Applied StandardScaler normalization; addressed class imbalance using class-weight balancing to improve sensitivity to failure cases
- Developed interpretable Logistic Regression model (ROC-AUC: 0.88) with coefficient analysis for stakeholder explainability
- Built GCP AutoML model (ROC-AUC: 0.92) as production model; evaluated using precision, recall, F1-score, and threshold tuning based on business cost constraints
- Deployed on Vertex AI for batch scoring and real-time inference via API; automated model retraining, monitoring, and drift alerting via Cloud Composer

**Project 2: ESG Analytics Platform | Banking Client – CI/CD Pipeline** | GCP · BigQuery · Dataflow · Dataproc · Cloud Composer · Spark Scala · Python · Docker · Kubernetes

- Developed and optimized end-to-end ETL pipelines on GCP using Dataflow (Apache Beam), enabling efficient large-scale data ingestion from Cloud Storage into BigQuery with significantly reduced execution time

- Migrated and transformed 20,000+ PySpark jobs to Spark Scala on Cloud Dataproc and optimized 20,000+ BigQuery views, improving query performance, maintainability, and downstream data accessibility
- Designed scalable BigQuery data models and transformation frameworks ensuring data consistency, integrity, and lineage across platforms and analytics layers
- Automated batch processing and workflow orchestration using Apache Airflow, reducing deployment failures and improving pipeline reliability
- Authored HLD and LLD design documents for GCP data architecture; developed Python automation scripts for seamless deployment workflows
- Optimized BigQuery queries using partitioning and clustering strategies, improving query performance by ~40% and reducing slot consumption
- Implemented monitoring and alerting using Cloud Monitoring and Cloud Logging for proactive tracking of pipeline health and latency
- Built ETL pipelines using Dataflow reducing data processing time by ~40%; delivered clean, tested data to analytics teams improving productivity by ~25%

### **Project 3: AI-Powered Resume Intelligence System Using RAG and Semantic Retrieval | GenAI R&D | Python**

· FAISS, Sentence Transformers, Cross-encoder

- Built an AI-powered Resume Intelligence System using RAG, FAISS, Sentence Transformers, and cross-encoder reranking, improving semantic recruiter search relevance by ~40% and retrieval accuracy from ~55% → ~80% compared to the baseline single-embedding architecture, validated using RAGAS evaluation metrics, while maintaining average query latency of ~1.5–2 seconds.
- Redesigned traditional resume parsing architecture from full-resume embeddings to section-wise semantic embeddings, improving answer relevancy scores from 3.1 → 4.4/5, reducing noisy candidate retrievals, and lowering retrieval latency through metadata-aware pre-filtering.
- Engineered a multi-stage retrieval pipeline with metadata filtering, FAISS vector search, and cross-encoder reranking, reducing retrieval latency by ~35% while significantly improving context precision, faithfulness, and semantic ranking quality over vector-only retrieval systems.
- Implemented a scalable RAG-based recruiter search engine using semantic chunking and intelligent reranking, enabling accurate real-time recruiter queries across skills, location, and experience with sub-2-second response latency.
- Developed a production-grade AI hiring intelligence platform with optimized embedding-space architecture, significantly improving semantic retrieval quality, contextual grounding, and candidate ranking performance compared to conventional keyword-based and full-document embedding approaches

**Research Associate – AI & Data Science | Amrita Centre for Wireless Networks & Applications | Aug 2014 – Apr 2019**

### **Project 1: Breast Cancer Recurrence Risk Prediction | Healthcare AI | Python · XGBoost · Random Forest · SHAP**

· FAISS · OpenAI Embeddings · RAG

- Developed end-to-end XGBoost ML solution to predict breast cancer recurrence risk on clinical and protein expression data from 2,000+ patients (Amrita Institute of Medical Sciences)
- Applied Random Forest-based feature selection (impurity-based importance), feature standardization, and weighted sampling for class imbalance
- Used SHAP analysis to interpret predictions, identify key clinical drivers, and mitigate feature importance bias
- Built RAG system using OpenAI embeddings and FAISS to summarize clinical scan reports, improving accessibility of medical insights
- Evaluated using Accuracy, Specificity, F1-Score, ROC-AUC; reported findings in collaboration with clinical research team

### **Project 2: Behavioural Modelling for Smart Building Applications | Python · Markov Models · Sensor Data · Time-Series Feature Engineering**

- Developed spatio-temporal behavioral prediction system using Markov Model for smart building energy optimization
- Engineered time-based and spatial features from sensor data to model occupancy dynamics; improved energy efficiency while preserving occupant comfort
- Strong foundation in data structures and algorithms (arrays, stacks, queues, trees, graphs) with solid understanding of time complexity and recursion
- Implemented Availability based Algorithm which helps to control electricity usage within a stipulated limit.
- Built multiple AI/ML proof-of-concepts (POCs), including chatbot systems, focusing on scalable backend integration and real-world solution deployment

## EDUCATION

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**M.Tech – Computer Science (Machine Learning)**, Amrita Vishwa Vidhyapeetham | 2012–2014

**B.Tech – Information Technology**, LBS College of Engineering, Kasaragod | 2008–2012

## CERTIFICATIONS & AWARDS

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- Infosys Certified Generative AI Professional – Foundation (June 2025)
- Infosys Certified ML Professional (June 2025)
- Infosys RISE INSTA Award (2023)
- GATE Scholarship (2012)
- Best Paper Award (2017)

## PUBLICATIONS & RESEARCH

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- Using CPS Enabled Micro Grid System for Optimal Power Utilization and Supply Strategy – Energy and Buildings, 2017
- A Smart Positioning System for Personalized Energy Management in Buildings – IEEE WiSPNET
- User Centered Energy Management Scheme for Smart Buildings – IEEE ICCACI 2017
- Parallel Sparse Coding for Categorical Data – Elsevier, ERCICA-14

## DECLARATION

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I hereby declare that the information provided above is true and accurate to the best of my knowledge and belief.

Place: Dubai

Athira Krishna

Date :